PHYS 4041/5041
Fall 2023
Homework 6
Due: In class, Thursday, October 26, 2023

1. If $f(x)$ and $g(x)$ are expanded in Legendre polynomials over $-1 \leq x \leq 1$, i.e.,

$$\begin{aligned} f(x) &= \sum_{l=0}^{\infty} a_l P_l(x) \quad \text{and} \\ g(x) &= \sum_{l=0}^{\infty} b_l P_l(x), \end{aligned}$$

show that

$$\int_{-1}^{1} f(x)\, g(x)\, dx = \sum_{l=0}^{\infty} \frac{2 a_l b_l}{2l+1},$$

which is Parseval's Theorem for Legendre series.

2. Determine the equilibrium temperature distribution $\Psi(r, \theta)$ inside a uniform sphere of radius R, assuming the temperature on the surface of the sphere is given by

$$\Psi(R, \theta) = T_0 + T_1 \cos^2 \theta,$$

where T_0 and T_1 are constants.

3. The generating function for the Legendre Polynomials $P_l(x)$ is

$$\Phi(x, h) = (1 - 2xh + h^2)^{-1/2}.$$

(a) Show that $(x-h)\frac{\partial \Phi}{\partial x} = h \frac{\partial \Phi}{\partial h}$.

(b) Use this fact to prove $x P_l'(x) - P_{l-1}'(x) = l P_l(x)$, where $l \geq 1$.

4. Determine the Legendre Polynomial expansion of

$$f(x) = \begin{cases} 1 & x \leq a \\ 0 & x > a \end{cases},$$

where a is a constant with $-1 < a < 1$. Evaluate any integrals you may encounter with the help of recurrence formulas.

5. Consider the Legendre differential equation with $\ell = 0$.
   (a) Show that the function

$$Q_0(x) = \frac{1}{2} \log \left(\frac{1+x}{1-x} \right)$$

   is a solution to the Legendre ODE with $\ell = 0$.
   (b) When using the Frobenus method with $s = 0$ and $\ell = 0$, show that the series consisting of only *odd* terms leads to the $Q_0(x)$ defined above.

6. The addition theorem for spherical harmonics states that

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y^*_{lm}(\theta,\phi) Y_{lm}(\theta',\phi'),$$

where

$$\begin{aligned} \cos\gamma &= \hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{r}}', \\ \hat{\boldsymbol{r}} &= \sin\theta\cos\phi\hat{\boldsymbol{x}} + \sin\theta\sin\phi\hat{\boldsymbol{y}} + \cos\theta\hat{\boldsymbol{z}}, \quad \text{and} \\ \hat{\boldsymbol{r}}' &= \sin\theta'\cos\phi'\hat{\boldsymbol{x}} + \sin\theta'\sin\phi'\hat{\boldsymbol{y}} + \cos\theta'\hat{\boldsymbol{z}}. \end{aligned}$$

Rather than ask you to prove this, I just let you exercise the formula a little bit:
   (a) Show that $\cos\gamma = \sin\theta\sin\theta'\cos(\phi-\phi') + \cos\theta\cos\theta'$.
   (b) What does the addition theorem mean when $l = 0$? This should be trivial.
   (c) What about when $l = 1$? Does this make sense?
   (d) What is $Y_{lm}(0,\phi)$? For what m values is it non-zero?
   (e) What happens to the addition theorem when $\theta' = 0$?